



Grade 6
General
Mathematics



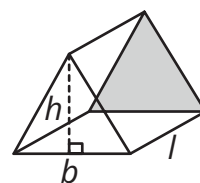
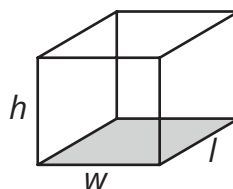
NSCAS NEBRASKA STUDENT-CENTERED ASSESSMENT SYSTEM

Mathematics Reference Sheet

2-Dimensional Shape	Area	Circumference
Circle	$A = \pi r^2$	$C = \pi d = 2\pi r$
Triangle	$A = \frac{1}{2}bh$	Perimeter
Rectangle	$A = l \times w$	$P = 2l + 2w = 2(l + w)$
Square	$A = s \times s$	$P = s + s + s + s$
Trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$	
Parallelogram	$A = bh$	

3-Dimensional Shape	Volume	Surface Area
Rectangular Prism	$V = lwh = Bh$	SA = the sum of the areas of all the faces
Triangular Prism	$V = \frac{1}{2}bhl = Bl$	SA = the sum of the areas of all the faces
Cone	$V = \frac{1}{3}\pi r^2h$	
Cylinder	$V = \pi r^2h$	
Sphere	$V = \frac{4}{3}\pi r^3$	

In the images to the right, the shaded faces are the bases.



2-Dimensional Shape Key		
b = base length	l = length	s = side
h = height	w = width	
d = diameter	r = radius	

Use 3.14 for π .

3-Dimensional Shape Key		
d = diameter	l = length	w = width
r = radius	h = height of shape	B = area of base
	b = base length	

Percent Change	Pythagorean Theorem
% change = $\frac{\text{difference in amounts}}{\text{original amount}}$	$c^2 = a^2 + b^2$

Standard Units	Metric Units
Conversions – Length	
1 foot (ft) = 12 inches (in.)	1 centimeter (cm) = 10 millimeters (mm)
1 yard (yd) = 3 feet (ft) = 36 inches (in.)	1 meter (m) = 100 centimeters (cm)
1 mile (mi) = 1,760 yards (yd) = 5,280 feet (ft)	1 meter (m) = 1,000 millimeters (mm)
	1 kilometer (km) = 1,000 meters (m)
Conversions – Volume	
1 cup = 8 fluid ounces (fl oz)	1 liter (l) = 1,000 milliliters (ml)
1 pint (pt) = 2 cups	1 liter (l) = 1,000 cubic centimeters (cu cm)
1 quart (qt) = 2 pints (pt)	
1 gallon (gal) = 4 quarts (qt)	
Conversions – Weight/Mass	
1 pound (lb) = 16 ounces (oz)	1 gram (g) = 1,000 milligrams (mg)
1 ton = 2,000 pounds (lb)	1 kilogram (kg) = 1,000 grams (g)

Directions:

On the following pages of your practice test booklet are questions for the Grade 6 *Nebraska Student-Centered Assessment System (NSCAS) General Mathematics* practice test.

Read these directions carefully before beginning the test.

This practice test will include several different types of questions. Multiple choice questions will ask you to select an answer from among four choices. Multiple select questions will ask you to select multiple correct answers from among five or more answer choices. For some questions, there may be two parts, Part A and Part B, where each part has a multiple choice or multiple select question. Some questions will ask you to construct an answer by following the directions given.

For all questions:

- Read each question carefully and choose the best answer.
- You may use scratch paper to solve the problems.
- The Mathematics Reference Sheet is provided in the back of the test booklet. You may refer to this page at any time during the test.
- Be sure to answer ALL the questions.

NSCAS GENERAL — MATHEMATICS

1. What is the value of y in the equation $\frac{y}{4} = 8$?

- A. 2
- B. 4
- C. 12
- D. 32

2. What is $\left| -\frac{2}{3} \right|$?

- A. $\frac{3}{2}$
- B. $-\frac{3}{2}$
- C. $\frac{2}{3}$
- D. $-\frac{2}{3}$

3. The table shows the relationship between x and y .

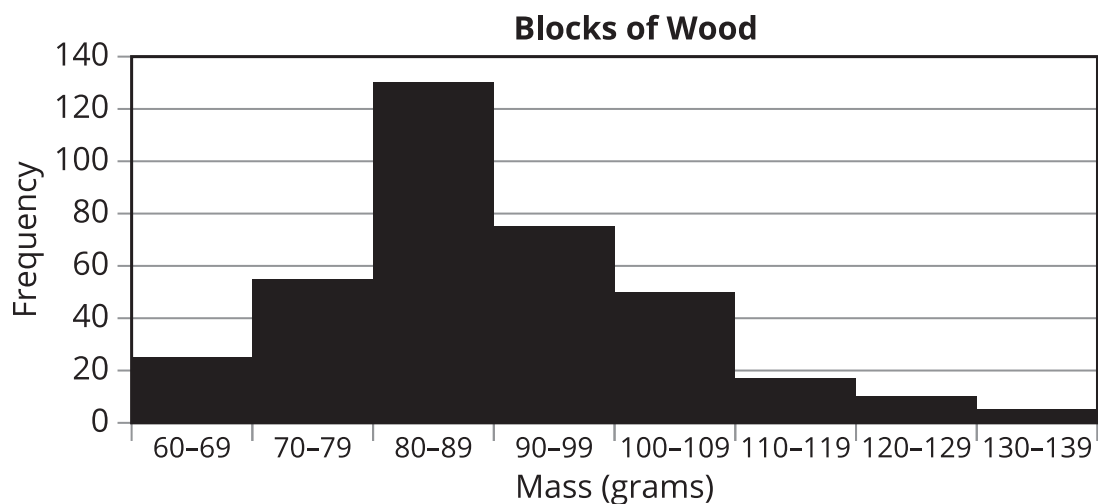
x	y
24	30
32	40
36	45
48	60
60	75

Which ordered pairs have the same relationship between x and y as the ordered pairs in the table?

Select **all** that apply.

- A. (16, 22)
- B. (44, 55)
- C. (68, 85)
- D. (75, 100)
- E. (160, 200)
- F. (180, 144)

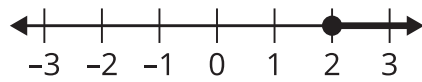
4. A carpenter has blocks of wood. The mass of the blocks of wood is shown in the histogram.



Which statement correctly describes the data?

- A. The data has an even distribution.
- B. The data distribution is skewed to the left.
- C. The data distribution is skewed to the right.
- D. The median of the data is between 100 and 109.

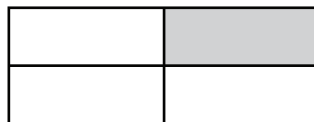
5. Use the number line to answer the question.



Which inequalities represent the solution shown on the number line?

Select **all** that apply.

- A. $4x \geq 8$
 - B. $x \leq 2$
 - C. $x - 1 \geq 1$
 - D. $x + 1 \leq 3$
 - E. $5x \geq 10$
 - F. $6x \geq 8$
6. Use the picture below to answer the question.

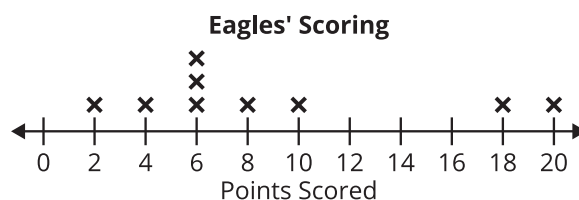
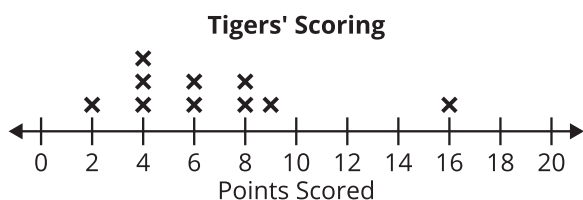


Which option lists the fraction, decimal, and percent that represent the shaded part?

- A. $\frac{1}{4}$, 0.20, 20%
- B. $\frac{1}{4}$, 0.25, 25%
- C. $\frac{1}{4}$, 0.4, 40%
- D. $\frac{1}{4}$, 0.75, 75%

NSCAS GENERAL — MATHEMATICS

7. The line plots show the number of points scored by players in the Tigers-Eagles basketball game.



Each **x** represents one player.

Based on the line plots, which statements are correct? Select true or false for each statement.

More Eagles players scored points than Tigers players.	True	False
The Eagles players scored more points than the Tigers players.	True	False
The median number of points scored per player was the same for both teams.	True	False
The mean number of points scored per player for the Tigers was greater than that for the Eagles.	True	False
The points scored by the Tigers players were more spread out than the points scored by the Eagles players.	True	False

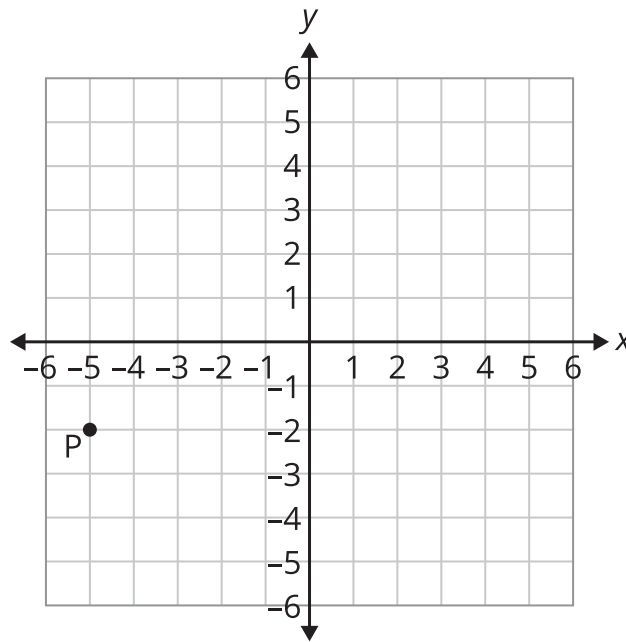
8. Which situation is BEST modeled by -15 ?

- A. growing 15 millimeters
- B. going 15 feet below sea level
- C. depositing \$15 into a savings account
- D. moving 15 steps forward

9. Which algebraic expression represents four times the quantity 22 less than x ?

- A. $4(x - 22)$
- B. $(4 \cdot 22) - x$
- C. $4(22 - x)$
- D. $(x \cdot 4) - 22$

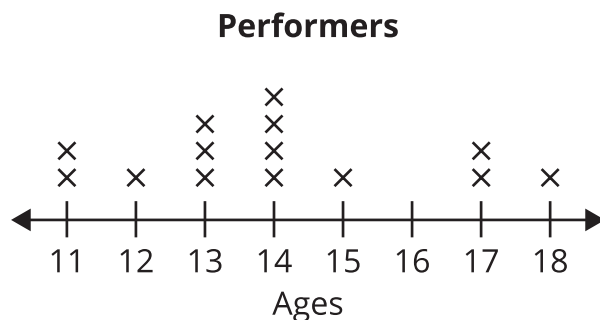
10. Use the graph to answer the question.



What is the ordered pair for point P? Write the answers in the boxes.

(,)

11. Use the line plot below to answer the question.



The line plot shows the ages of the performers in a play. How many performers are older than 13?

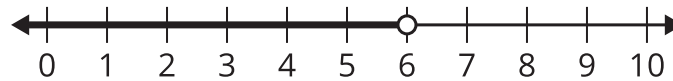
- A. 4 performers
- B. 8 performers
- C. 11 performers
- D. 14 performers

12. What is the quotient?

$$5.2 \div 10^2 = ?$$

- A. 0.052
- B. 0.52
- C. 520
- D. 5,200

13. Use the number line to answer the question.



Which inequality is represented by the number line? Write a symbol or number in each box.

< ≤ > ≥ 0 6

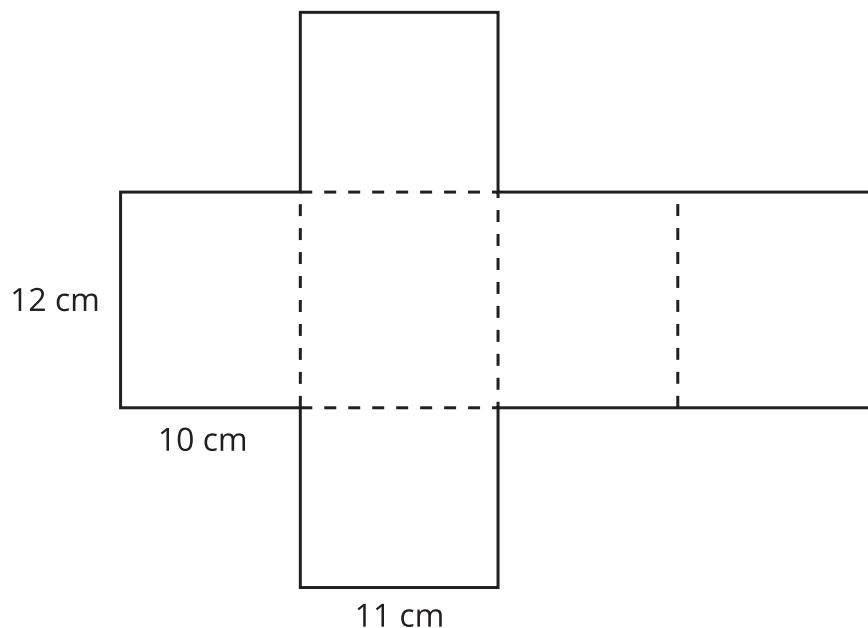
x

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14. Owen has $1\frac{1}{2}$ gallons of fruit punch. He pours the punch into glasses that hold $\frac{1}{16}$ gallon. How many glasses can Owen fill with fruit punch?
- A. 11 glasses
 - B. 24 glasses
 - C. 32 glasses
 - D. 48 glasses

15. This question has two parts. Answer Part A, and then answer Part B.

Use the net to answer the questions.



Part A

What are the areas of the different faces of the figure formed by the net?

Select **all** that apply.

- A. 100 square centimeters
- B. 110 square centimeters
- C. 120 square centimeters
- D. 121 square centimeters
- E. 132 square centimeters
- F. 144 square centimeters

Part B

What is the surface area of the figure formed by the net? Write the answer in the box.



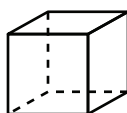
square centimeters

16. When $n = 11$, what is the value of $10 - (n - 6)$? Write the answer in the box.

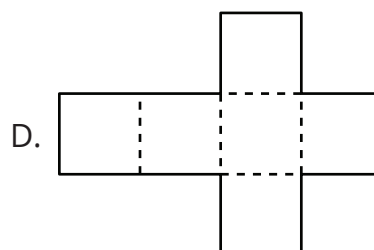
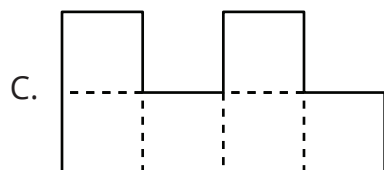
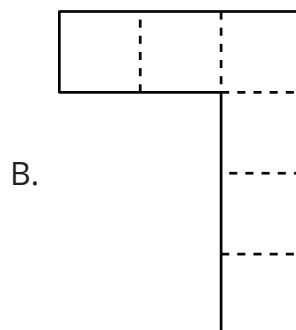
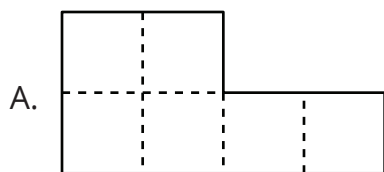
17. Which expression is the prime factorization of 100?

- A. $2^2 \cdot 5^2$
- B. $2^2 \cdot 25$
- C. $4 \cdot 5^2$
- D. $4 \cdot 25$

18. Use the figure to answer the question.



Which net could be used to form the figure?



19. Use the table to answer the question.

Number of Juice Boxes	6	12	18	24
Number of Cookies	10	20	30	40

Which ratios of juice boxes to cookies are equivalent to the ratios in the table?

Select **all** that apply.

- A. $\frac{1 \text{ juice box}}{2 \text{ cookies}}$
- B. $\frac{9 \text{ juice boxes}}{15 \text{ cookies}}$
- C. $\frac{34 \text{ juice boxes}}{50 \text{ cookies}}$
- D. $\frac{36 \text{ juice boxes}}{60 \text{ cookies}}$
- E. 3 juice boxes to 5 cookies
- F. 15 juice boxes to 9 cookies
- G. 21 juice boxes to 35 cookies

20. Which value of r makes the equation $115 = 5r$ true?

- A. $r = 23$
- B. $r = 28$
- C. $r = 110$
- D. $r = 115$

**NSCAS General Grade 6
Practice Test Answer Key
Mathematics (English)**



Sequence	Key	Points
1	D	1
2	C	1
3	B, C, E	2
	2 of 3 correct responses	1
4	C	1
5	A, C, E	2
	2 correct, 0 incorrect	1
	3 correct, 1 incorrect	1
6	B	1
7	False True True False False	2
	4 correct, 1 incorrect	1
	4 correct, 0 incorrect	1
	3 correct, 2 incorrect	1
	3 correct, 1 incorrect	1
	3 correct, 0 incorrect	1
8	B	1
9	A	1
10	-5, -2	1
11	B	1
12	A	1
13	$x \begin{array}{ c } \hline < \\ \hline \end{array} \begin{array}{ c } \hline 6 \\ \hline \end{array}$	1
14	B	1
15	Part A: B, C, E Part B: 724 or equivalent	2
	Part A or Part B	1
16	5 or equivalent	1
17	A	1
18	D	1
19	B, D, E, G	2
	3 correct, 1 incorrect	1
	3 correct, 0 incorrect	1
	2 correct, 0 incorrect	1
20	A	1